

Literature Review: multimodal fake news classification

1 Introduction

In today’s world, the spread of misinformation on social media is posing a crucial challenge, as deceptive content is increasingly combining textual and visual modalities to seem more credible. In this situation, traditional text-only detection systems fail, especially when images are used to manipulate emotions, mislead readers, or provide false context. For example, authentic images may be paired with fabricated captions, or visuals may be altered to align with misleading narratives. This fusion of modalities makes misinformation more persuasive and harder to detect.

Recent advances in deep learning offer promising ways to tackle this. Large language models (LLMs) can capture subtle semantic and linguistic patterns, while convolutional neural networks (CNNs) excel at identifying visual artifacts and stylistic cues. However, combining these signals and using them to identify fake news remains a major problem. Recent works have tried to build multimodal misinformation detection systems that process textual and visual features to classify news as real or fake. By using cross-modal attention and representation learning, these methods seek to enhance robustness against content manipulation and improve generalizability of fake news detection systems.

2 Problem Settings and Datasets

Multimodal fake news detection can be viewed as binary or multi-class classification task where both text and images are used to predict labels. Unlike unimodal approaches, this setting requires fusion strategies to align and integrate textual semantics with visual features. Some methods focus on early fusion (concatenating features before classification) while others use late fusion (combining modality-specific predictions). In addition, cross-modal inconsistency detection aims to identify mismatches between textual claims and accompanying visuals, while visual grounding approaches evaluate whether image regions support or contradict textual assertions. These problem formulations require careful handling of multimodal noise, missing modalities, and domain shifts.

2.1 Key Datasets

Several benchmark datasets support multimodal misinformation research: Fakeddit — over 1M Reddit posts with titles, images, and metadata labeled at 2/3/6-way veracity levels; widely used for multimodal classification [1]. Weibo — social media posts with text-image pairs labeled as fake or real, often used for cross-modal inconsistency detection [2]. Twitter MediaEval Verifying Multimedia Use — event-based dataset with tweets, attached images, and labels indicating if media is reused out of context [3]. FakeNewsNet — combines textual content, publisher metadata, and user engagement graphs, including some posts with images [4]. LIAR (text-only) — included as a baseline for language-only comparison [15]. These datasets differ in scale, modality coverage, and label granularity, collectively enabling evaluation of multimodal models across both content-based and consistency-based detection paradigms.

3 Methods by Signal Type

Research on fake news detection has explored a variety of modeling paradigms depending on the **input signals** used—textual content, social context, and multimodal (text+image) information. Each signal provides complementary cues for identifying deceptive content, and integrating them can improve robustness.

3.1 Text-Based Methods

Early work in fake news detection relied on traditional *shallow features* such as bag-of-words, n-grams, and TF-IDF representations, fed into classifiers like SVMs, logistic regression, and random forests. While these models were interpretable, they struggled to capture semantic nuances and contextual dependencies.

With the advent of pre-trained Transformer models, text-based methods improved dramatically. Fine-tuning BERT, RoBERTa, and DeBERTa on domain-specific fake news corpora has become standard, enabling models to detect subtle semantic shifts, hedging language, and affective tone often found in deceptive content.

Recent methods also incorporate *stance detection and evidence retrieval* to improve generalization across topics. For example, systems trained on the FEVER dataset learn to verify claims against retrieved Wikipedia evidence [5], improving their ability to judge unfamiliar claims. However, these retrieval-based systems remain underused in fake news detection, partly due to the high cost of constructing and maintaining knowledge bases for dynamic social media environments.

Overall, text-based models offer strong baselines, but they are vulnerable to manipulated visuals and often fail when textual content is sparse or neutral yet paired with misleading images.

3.2 Social Context and Propagation-Based Methods

Text alone rarely captures the full dynamics of misinformation spread. Research has shown that *user engagement and propagation patterns* are strong indicators of credibility. Fake news often spreads faster, deeper, and more broadly than real news.

Graph-based approaches model these dynamics explicitly. For example, Bi-GCN (Bipartite Graph Convolutional Network) builds a graph with users and news articles as nodes, passing messages between them to learn credibility representations [10]. UPFD (User Preference-aware Fake News Detection) similarly uses Graph Neural Networks (GNNs) to combine article features with user preference embeddings, showing strong performance when early propagation data is available [11].

Other methods model conversation trees from Twitter (e.g., PHEME) using tree-structured GCNs or hierarchical attention networks to detect misinformation based on the structure and sentiment of replies [6]. These approaches often outperform text-only models in early detection scenarios, though they rely on having rich and timely social signals, which may not always be accessible.

3.3 Multimodal (Text + Image) Methods

As misinformation increasingly pairs text with images to enhance perceived credibility, *multimodal detection methods* have become essential. These models aim to learn joint representations that capture the semantic alignment—or misalignment—between text and visual content.

EANN (Event-Adversarial Neural Network) introduced an adversarial training scheme to learn event-invariant features across text and image modalities, reducing overfitting to specific news events [7]. SpotFake and SpotFake+ combine BERT-based text encoders with CNN-based image encoders, fusing the two modalities via late concatenation before classification [8]. SAFE (Similarity Aware Fake news Detection) goes further by explicitly modeling cross-modal similarity, penalizing mismatched text–image pairs to detect repurposed or misleading visuals [9].

More recent work leverages CLIP (Contrastive Language–Image Pretraining), which aligns text and images in a shared embedding space, enabling zero-shot or few-shot fake news detection [14]. CLIP-based models are especially promising for open-domain settings, where misinformation topics shift rapidly and labeled data is scarce.

Multimodal systems still face challenges such as missing modalities, noisy OCR text in memes, and the need for effective fusion architectures (early, late, or hybrid). However, they represent the **state-of-the-art direction** for combating modern, visually-enhanced misinformation, and are the primary focus of this project.

4 State-of-the-Art (SOTA) Baselines

Recent work has established strong baselines for fake news detection across text, graph, and multimodal signals.

4.1 Text-Based

Transformer-based models dominate text-only detection. RoBERTa and DeBERTa, fine-tuned on headline–body pairs, outperform CNN/RNN models by capturing subtle linguistic cues like hedging and stance shifts. Domain adversarial training improves cross-topic generalization by forcing the model to learn source-invariant features [12]. Some pipelines also integrate stance/evidence retrieval (e.g., FEVER) to verify claims against factual sources, further boosting performance over classical TF–IDF+SVM baselines.

4.2 Graph-Based

Graph neural networks (GNNs) leverage user–news propagation patterns, which are highly predictive. Bi-GCN models user–news bipartite graphs and jointly learns user credibility and news veracity [10]. GAT and GraphSAGE, evaluated on the UPFD benchmark, use attention or neighborhood sampling to capture relational signals [11]. These models often surpass text-only baselines when early social engagement data is available.

4.3 Multimodal

Multimodal models fuse text and image signals and set SOTA on datasets like Fakeddit. EANN introduces event-adversarial loss to learn event-invariant features [7]. SpotFake+ combines BERT (text) and CNN (image) streams via late fusion [8], while SAFE aligns text–image pairs to catch semantic mismatches [9]. More recently,

CLIP-based vision–language models have achieved strong results through shared embedding spaces and cross-attention fusion [14], showing robust generalization to unseen topics.

Overall, these baselines represent the SOTA: text-based transformers excel in language modeling, GNNs capture social dynamics, and multimodal CLIP-based systems offer the strongest performance when both modalities are available.

5 Research Gaps and Future Directions

Despite progress in text, graph, and multimodal methods, several key gaps limit current fake news detection systems:

(1) **Domain Shift.** Many models overfit to specific publishers, topics, or events, leading to sharp performance drops on unseen content. Domain-adversarial training and test-time adaptation are underexplored solutions [7].

(2) **Cross-Modal Inconsistency.** Most multimodal models focus on feature fusion but rarely assess whether the text and image actually agree. Detecting out-of-context or repurposed visuals remains challenging [9].

(3) **Limited Data for Emerging Events.** Labeled data is scarce for new misinformation topics. Few-shot learning, data augmentation, and retrieval-augmented models could improve adaptability [14].

(4) **Dependence on Social Graphs.** GNN-based methods require user–content interaction graphs, which are often unavailable due to privacy or platform restrictions. More work on content-only architectures is needed.

(5) **Lack of Explainability.** Current models act as black boxes, offering little insight into their decisions. Generating interpretable rationales or highlighting contradictory evidence could increase user trust [11].

(6) **Multilingual and Noisy Data.** Most datasets are English-only, while real-world misinformation spans languages and scripts. Robust multilingual models and noise-resistant preprocessing are still rare [13].

Addressing these gaps will be critical for building robust and generalizable multimodal misinformation detection systems.

6 Conclusion

Fake news detection has evolved from simple text classifiers to sophisticated multimodal and graph-based systems. Text-based transformers like RoBERTa and DeBERTa offer strong semantic modeling, GNNs leverage social propagation cues, and multimodal models such as EANN, SpotFake+, SAFE, and CLIP achieve the best results when both text and images are available.

However, challenges like domain shift, cross-modal inconsistency, and data scarcity persist. Future systems must combine robust content modeling with cross-modal reasoning, adapt to new events with minimal data, and provide interpretable outputs. Advancing in these directions is essential to make automated misinformation detection both accurate and trustworthy.

References

- [1] Nakamura, K. et al. (2019). *Fakeddit: A new multimodal benchmark dataset for fine-grained fake news detection*. arXiv:1911.03854
- [2] Jin, Z. et al. (2017). *Multimodal fusion for fake news detection on social media*. ICPR

- [3] Boididou, C. et al. (2018). *Verifying Multimedia Use at MediaEval 2018*. MediaEval Workshop
- [4] Cao, Q. et al. (2018). *FakeNewsNet: A Data Repository with News Content, Social Context and Dynamic Information*. arXiv:1809.01286
- [5] Thorne, J. et al. (2018). *FEVER: A large-scale dataset for fact extraction and verification*. NAACL
- [6] Zubiaga, A. et al. (2018). *Detection and resolution of rumours in social media: A survey*. ACM CSUR
- [7] Wang, Y. et al. (2018). *EANN: Event Adversarial Neural Networks for Multi-modal Fake News Detection*. KDD
- [8] Khattar, D. et al. (2019). *SpotFake: A multimodal variational autoencoder for fake news*. arXiv:1907.08155
- [9] Zhou, L. et al. (2020). *SAFE: Similarity-Aware Multi-Modal Fake News Detection*. arXiv:2003.04981
- [10] Bian, T. et al. (2020). *Rumor Detection with Bi-Directional GCNs*. AAAI Workshop
- [11] Dou, Y. et al. (2021). *User Preference-Aware Fake News Detection with GNNs*. KDD Workshop
- [12] Zhou, X., Zafarani, R. (2020). *A Survey of Fake News: Theories, Detection Methods, and Opportunities*. ACM CSUR
- [13] Srivastava, A. et al. (2023). *A Survey of Fake News Detection using Deep Learning*. arXiv:2310.03408
- [14] Radford, A. et al. (2021). *Learning Transferable Visual Models From Natural Language Supervision*. ICML
- [15] Wang, W.Y. (2017). *“Liar, Liar Pants on Fire”: A New Benchmark Dataset for Fake News Detection*. ACL